

Gate A — Answer Sheet

Instructions

- Areas marked `<insert here>` are where you write your answers.
 - Areas marked `![screenshot]` are where you paste screenshots.
 - **Tip — pasting screenshots:** Use the [Paste Image](#) VS Code extension (Built into container).
 - **Tip — export to PDF:** Use the [Markdown PDF](#) VS Code extension. Export and submit the PDF to Brightspace (Built into container).
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Student Info

Name: Job bouwhuis Class: Sb Teacher: Melanie Bonnes

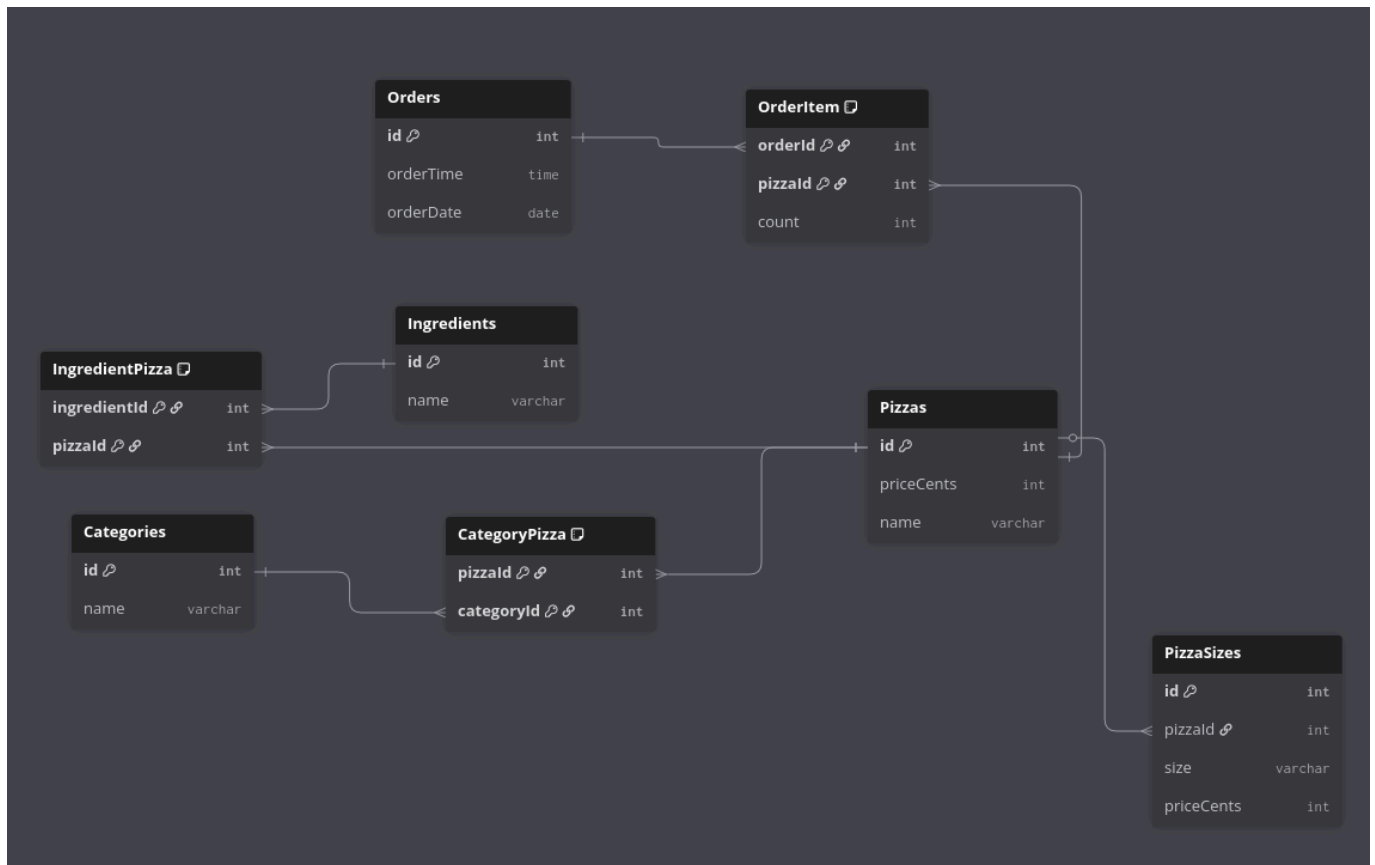
Part 1 — Pizza Normalization

Exercise: `student-preload` → `gatekeepers/gate-a-pizza`

Improved ERD

Design your improved schema in dbdiagram.io starting from the provided `pizza.dbml`.

Screenshot of your improved ERD:



Justification

Explain your design choices in your **own words** . For each change you made, answer:

- What did you change?
- Why did you change it? (What problem does it solve?)

it was one table with lots of data that belongs to different data models, now its multiple tables so that data can be structured once, and therefor no duplicate data exists (unless i missed something)

Part 2 — StackExchange Optimization

Exercise: [student-preload](#) → [gatekeepers/gate-a-stackexchange](#)

Part 2.1 — Baseline Performance

Query 1 — Search

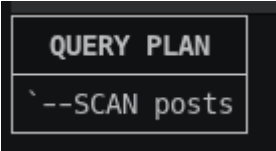
Find questions or answers that mention the word **postgres** in the title or body.

SQL:

```
SELECT *  
FROM posts
```

```
WHERE title LIKE '%postgres%'
OR body LIKE '%postgres%';
```

Explanation (in your own words): select everything from posts where title or body contains "postgres" anywhere in them



Screenshot — Query Plan:

Screenshot — First 3 Records:  alt text

Query 2 — Questions Overview

Show all questions sorted by date descending, including the first 200 characters of the content, the number of answers, and the username of the author.

SQL:

```
SELECT
  post.title,
  SUBSTR(post.body, 1, 200) AS preview,
  usr.displayname,
  COUNT(cmts.id) AS answers
FROM Posts post
LEFT JOIN Comments cmts ON cmts.postid = post.id
JOIN Users usr ON usr.id = post.owneruserid
ORDER BY post.creationdate DESC;
```

Explanation (in your own words): selecting the title the first at most 200 characters of the body from posts joining comments to get the number of answers joining on user to get the poster displayname and ordering by post date descending

Screenshot — Query Plan:

	abc id ▾ ⋮	abc parent ▾ ⋮	abc notused ▾ ⋮	abc detail ▾ ⋮
1	11	0	194	SCAN usr
2	13	0	52	SEARCH post USING INDEX idx_posts_owneruserid (owneruserid=?)
3	20	0	51	SEARCH cmts USING INDEX idx_comments_postid (postid=?) LEFT-JOIN
4	29	0	0	USE TEMP B-TREE FOR GROUP BY
5	83	0	0	USE TEMP B-TREE FOR ORDER BY

Screenshot — First 3 Records:

	abc title ▾ ⋮	abc preview ▾ ⋮	abc displayname ▾ ⋮	abc answers
1	Inserting Rows withi...	<p>I am working with Netezza S...	stats_noob	0
2	aurora mysql - migra...	<p>We have write intensive app...	JollyRoger	0
3	(null)	<p>The following was the answe...	MJM	0
4	Subtracting All Valu...	<p>I am trying to subtract all...	MJM	0
5	(null)	<pre><code> select i,nextval('...	jjanes	0
6	Alerting system rela...	<p>I'm no expert in DBMS and t...	Laurent	0
7	(null)	<p>The results of a query are ...	Charlieface	0
8	(null)	<blockquote> <p>What I'm now l...	Paul White	1
9	Unable to import mor...	<p>I have configured NDBCLUSTE...	Debabrata	0
10	(null)	<p>Your function (not stored p...	jjanes	0
11	(null)	Use this: <pre><code>de...	Vérace	1

Query 3 — Insert Answer

Insert a new answer for an existing question.

SQL:

```
INSERT INTO Comments (postid, userid, text)
VALUES (
  733,
  171,
  'This was inserted!!!'
);
```

Explanation (in your own words): there isnt much to explain here.....

Part 2.2 — Optimize

Indexes

SQL to create indexes:

```
CREATE INDEX idx_posts_title ON Posts(title);

CREATE INDEX idx_posts_body ON Posts(body);

CREATE INDEX idx_comments_body ON Comments(body);
```

Explanation — why these columns, why these index types: used for text queries, such as the earlier %postgres%

Denormalization & Triggers

SQL — Alter table (add denormalized column):

```
ALTER TABLE Posts
ADD answercount INTEGER DEFAULT 0;
```

SQL — Populate the new column:

```
UPDATE Posts
SET answercount = (
    SELECT COUNT(*)
    FROM Comments
    WHERE Comments.postid = Posts.id
);
```

SQL — Trigger(s) to keep it in sync:

```
CREATE TRIGGER trg_comments_insert
AFTER INSERT ON Comments
BEGIN
    UPDATE Posts
    SET answercount = answercount + 1
    WHERE id = NEW.postid;
END;

CREATE TRIGGER trg_comments_delete
AFTER DELETE ON Comments
BEGIN
    UPDATE Posts
    SET answercount = answercount - 1
    WHERE id = OLD.postid;
END;
```

Explanation — what you denormalized and why: now instead of having to join comments on their postid back to the current post, and counting the amount of comments we have, its just a column that is kept in sync.

Part 2.3 — Compare Results

Query 1 — Search (Optimized)

SQL:

```
SELECT *
FROM posts
```

```
WHERE title LIKE '%postgres%'
OR body LIKE '%postgres%';
```

Explanation — what changed and what impact did it have: it was faster, thats generally what indexing does

Screenshot — Query Plan:

	# id ▾ ⚙	# posttypeid ▾ ⚙	# parentid ▾ ⚙	# acceptedan
1	11916	2	11893	(null)
2	12506	2	12501	(null)
3	78398	2	78350	(null)

Screenshot — First 3 Records:

	abc title ▾ ⚙	abc preview ▾ ⚙	abc displayname ▾ ⚙	# answercount ▾ ⚙
1	Inserting Rows within a CTE	<p>I am working with Ne...	stats_noob	0
2	aurora mysql - migrating large tables	<p>We have write intens...	JollyRoger	0
3	(null)	<p>The following was th...	MJM	0
4	Subtracting All Values of Column by Each ...	<p>I am trying to subtr...	MJM	0
5	(null)	<pre><code> select i,ne...	jjanes	0
6	Alerting system relationships and STI in ...	<p>I'm no expert in DBM...	Laurent	0
7	(null)	<p>The results of a que...	Charlieface	0
8	(null)	<blockquote> <p>What I'...	Paul White	1
9	Unable to import more than 110 Columns in...	<p>I have configured ND...	Debabrata	0
10	(null)	<p>Your function (not s...	jjanes	0
11	(null)	Use this: <pre><...	Vérace	1
12	Country, City and Airport data query	<p>I have a csv/excel f...	Java_Beginner	0
13	(null)	Use this: <pre><...	Vérace	0
14	SQL: How to look within each "cluster" va...	<p>I have a scenario wh...	Peter Danzo	0
15	Sequence does skip numbers	<p>I have a sequence li...	Luuk	1
16	Different Spool usage for running the sam...	<p>How can spool usage ...	Mohamed Fathi	0
17	Postgresql: Performance - SELECT with fun...	<p>On Postgrsql 13.5 - ...	bboness	1
18	Why does SELECT FOR UPDATE not support ag...	<p>I checked the Postgr...	Mahatma_Fatal_Error	1
19	Modifying PostgreSQL Table for Handling M...	<p>I'm working on a pro...	Maine	0
20	migrated from mysql5.7 to mysql8 - percon...	<p>I have just migrated...	clarkk	0

Query 2 — Questions Overview (Optimized)

SQL:

```
SELECT
  post.title,
  SUBSTR(post.body, 1, 200) AS preview,
  usr.displayname,
  post.answercount
FROM Posts post
JOIN Users usr ON usr.id = post.owneruserid
ORDER BY post.creationdate DESC;
```

Explanation — what changed and what impact did it have: removed the joining on comments stripping thousands of checks. making it faster

Screenshot — Query Plan:

1	5	0	194	SCAN usr
2	7	0	52	SEARCH post USING INDEX idx_posts_owneruserid (owneruserid=?)
3	24	0	0	USE TEMP B-TREE FOR ORDER BY

Screenshot — First 3 Records:

	abc title	abc preview	abc displayname	# answercount
1	Inserting Rows within a CTE	<p>I am working with Ne...	stats_noob	0
2	aurora mysql - migrating large tables	<p>We have write intens...	JollyRoger	0
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5	(null)	<pre><code> select i,ne...	jjanes	0
6	Alerting system relationships and STI in ...	<p>I'm no expert in DBM...	Laurent	0
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19	Modifying PostgreSQL Table for Handling M...	<p>I'm working on a pro...	Maine	0
20	migrated from mysql5.7 to mysql8 - percon...	<p>I have just migrated...	clarkk	0

Query 3 — Insert Answer (Optimized)

SQL:

```
INSERT INTO Comments (postid, userid, text)
VALUES (
    733,
    171,
    'This was inserted after indexing was added!'
);
```

Explanation — what changed and what impact did it have: still dont know what to explain here, it inserts the comment. only thing i can add now is that it upped the answercount on the post it was added to due to the trigger _(_)_/